# Worksheet 5 *Multipliers and exponents*

Prepares you for lab manual pages **Exp1, Exp2, Exp3, Exp4** (section L2 S4)

Name: Name of TA:

**1.** Fill in the blanks to find an equivalent multiplier.

(a) *Example:* Increasing $X$ by 50% is the same as multiplying: $1.5 \cdot X$.

(b) Decreasing $P$ by 20% is the same as multiplying: $\square \cdot P$.

(c) Increasing $N$ by 4% is the same as multiplying: $\square \cdot N$.

**2.** Exponents describe repeated multiplication.

(a) Write $1.5 \cdot 1.5 \cdot 1.5 \cdot 1.5$ as a power of 1.5.

(b) How many 20-minute periods are in 2 hours? In $t$ hours?

(c) Write $2^{-t}$ in the form $b^t$. $b =$

(d) Write $2^{t/10}$ in the form $b^t$. $b =$

**3.** Let $F(t) = 300 \cdot (1.2)^t$. Use a calculator.

(a) $F(0) =$ (b) $F(2) =$ (c) $F(2.5) =$